MILITARY SPECIFICATION

DEMOLITION KIT, BLASTING: XM268 (BLASTING AGENT)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a two component military slurry blasting agent (see 6.1), which is activated by mixing two parts of the liquid component with one part of the solid component.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation and Packaging, Methods of
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, A General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1376

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STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-650 - Explosives: Sampling, Inspection and Testing

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

MIL-STD-1218 - ACS Chemicals

DRAWINGS

US ARMY

9328450	- Demolition Kit, Blasting: XM268
9329679	- Container Assembly
9329739	- Tie, Bag
9331371	- Aluminum Powder
9331372	- Ethylene Glycol
9331373	- Sodium Perchlorate
9331374	- Guar Gum
9331375	- Isostearic Acid
9331376	- Potassium Acid Phthalate
11828915	- Packing and Marking for Box, Pallet, Wirebound For Demolition Kit, Blasting: XM268
11828917	- Packaging and Marking for Container Assembly
11828919	- Bag, Charge

(Copies of specifications, standards, drawings and publications, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATION

ASTM Designation E300-70 - Recommended Practice for Sampling Industrial Chemicals

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(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

3. REQUIREMENTS

3.1 Material. The materials used in the manufacture of the Demolition Kit, Blasting: XM268 (Blasting Agent) shall conform to the following drawings: Aluminum - drawing 9331371, ethylene glycol - drawing 9331372, sodium perchlorate - drawing 9331373, guar gum - drawing 9331374, isostearic acid - drawing 9331375 and potassium acid phthalate - drawing 9331376.

3.2 Composition. The blasting agent shall be supplied as two separate mixtures which shall conform to Table I when tested as described in 4.5.

TABLE I

Test Method	Ingredient	Percent (by weight)
	Liquid Component	
4.5.1.1.1	Sodium Perchlorate	51 ± 2.0
4.5.1.1.3	Water	39 ± 2.0
4.5.1.1.2	Ethylene Glycol	10 ± 2.0
	Acetic Acid to adjust pH to 4.6 ± 0.1	
	Powder Component	
4.5.2.1.1	Aluminum (Coated with isostearic acid)	97.6 ± 2.0
4.5.2.1.2	Guar Gum	1.4 ± 0.2
4.5.2.1.3	Potassium Acid Phthalate	1.0 ± 0.2

The blasting agent shall be activated by thoroughly mixing 64 ± 3% of the liquid component with 36 ± 3% of the powder component.

3.3 pH. The pH of the liquid component shall be 4.6 ± 0.1 when tested as described in 4.5.1.2.

3.4 Total volatiles. The total volatiles of the powder component shall be 0.08% (max) when tested as described in 4.5.2.2.

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3.5 Density. The specific gravity of the mixed blasting agent shall be a minimum of 1.45 when tested as specified in 4.5.3.1.

3.6 Viscosity. The viscosity of the mixed blasting agent shall be 6500 centipoises, min. when tested as specified in 4.5.3.2.

3.7 Functioning. The functioning test described in 4.5.3.3 shall be performed on the mixed blasting agent. The blasting agent shall be considered acceptable if complete detonation occurs.

3.8 Detonation velocity. The detonation velocity of the mixed blasting agent shall be determined as described in 4.5.3.3 for informational purposes.

3.9 Bullet impact sensitivity. The bullet impact sensitivity test described in 4.5.3.4 shall be performed on the mixed blasting agent. The blasting agent shall be considered acceptable if no detonations or burning occurs.

3.10 Workmanship. The composition of each component shall be formulated in a thorough workmanlike manner and all manufacturing and processing shall be free of contaminants. In addition, the manufacturer shall employ procedures which provide controls to preclude any condition which may adversely affect the safety or reliability of the blasting agent. The test shall be conducted as described in 4.5.4.

3.11 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (See 6.2)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection. (see 6.2)

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of one hundred pounds of the composition obtained by sampling as described in 4.4.3.1. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

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TABLE II - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liquid Component				
	Composition			3.2	4.5.1.1
	pH			3.3	4.5.1.2
	Powder Component				
	Composition			3.2	4.5.2.1
	Total Volatiles			3.4	4.5.2.2
	Mixed Blasting Agent				
	Density			3.5	4.5.3.1
	Viscosity			3.6	4.5.3.2
	Functioning			3.7	4.5.3.3
	Detonation velocity			3.8	4.5.3.3
	Blasting cap sensitivity			3.9	4.5.3.4
	Workmanship			3.10	4.5.4

NOTES:

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4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in Table II of this specification.

4.3.3 Rejection. See MIL-A-48078

4.4 Quality conformance inspection

4.4.1 Lot formation. A lot shall consist of one or more batches of blasting agent, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. A batch shall consist of that quantity of blasting agent that has been subjected to a single unit chemical or physical mixing process intended to make the final product homogeneous. Each individual component used in the mixes shall be supplied by a single manufacturer and produced by a single process.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (See MIL-A-48078).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63260 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Container Assembly, empty		9329679
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Holes, splits, cracks in bottles or caps		0.40%	5.3.1	Visual
102	Deformed - i.e. warped, twisted or irregular contours		0.40%	5.3.1	Visual
103	Excessively thin areas		0.40%	5.3.1	Visual/Manual
104	Less than minimum weight		0.40%	5.3.1	Scale
105	Land less than 1/8 inch wide or not smooth		0.40%	5.3.1	Visual/Gauge
106	Color, embossed bars and net weight incorrectly matched		0.40%	5.3.1	Visual
107	Cap missing or not corresponding to color of container		0.40%	5.3.1	Gauge
108	Contaminated - i.e. dirt, oil, loose particles or scraps		100%	5.3.1	Visual
MINOR					
201	Marking incorrect or illegible		0.65%	5.3.1	Visual
202	Sharp flash inside handle		0.65%	5.3.1	Visual
203	Scuffed or dented		0.65%	5.3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63260 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Container Assembly, filled				11828917
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Container leaks		100%	5.1.1	4.5.5/Visual
102	Container or cap punctured		100%	5.1.1	4.5.5/Visual
103	Container bulged		100%	5.3.1	Visual
104	Cap missing		100%	5.3.1	Visual
105	Container color not correct for contents		100%	5.1.1	Visual
MINOR					
201	Cap wrong color		0.65%	5.1.1	Visual
202	Container scuffed, dented or discolored		0.65%	5.1.1	Visual
203	Marking incorrect or illegible		0.65%	5.1.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63260 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Packaged items (goggles, gloves, blasting agent charge bags, charge bag ties, (technical manual) prior to packaging.		9328450/9329739/ 11828919
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
	Goggles				
101	Incorrect quantity		100%	5.1.4	Visual
102	Lenses damaged or dirty		100%	5.1.4	Visual
103	Strap damaged or missing		100%	5.1.4	Visual
	Rubber gloves				
104	Incorrect quantity or mismated		100%	5.1.5	Visual
105	Torn or punctured		100%	5.1.5	Visual
	Charge bags				
106	Incorrect quantity		100%	5.1.2	Visual
107	Wrong size		100%	5.1.2	Visual
108	Torn or punctured		100%	5.1.2	Visual
	Charge bag ties				
109	Incorrect quantity		100%	5.1.3	Visual
	Technical manual				
110	Manual missing		100%	5.1.6	Visual
MINOR	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63260 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Packaged Items (goggles, gloves, blasting agent charge bags with ties) and technical manual) after sealing	1 OF 1	11828915
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Barrier bag heat seal incorrect		0.40%	5.1	Visual
102	Holes, tears, punctures in barrier bag		0.40%	5.1	Visual
103	Packaged item (except goggles) more than 2 inches thick		0.40%	5.1	Gauge
MINOR					
201	Bag not tight against contents		0.65%	5.1	Visual
202	Marking incorrect or illegible		0.65%	5.3	Visual
203	Bag damaged but not punctured		0.65%	5.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63260 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11828915
4.4.2.5	Demolition Kit prior to closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
MAJOR					
101	Contents incorrect		0.40%	5.2	Visual
102	Protruding nails, staples or splinters		0.40%	5.2	Visual
MINOR					
201	Contents incorrectly arranged		0.65%	5.2	Visual
202	Dirt or debris present		0.65%	5.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63260 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11828915
4.4.2.6	Demolition Kit after closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Broken board		0.40%	5.2	Visual
102	Broken wires		0.40%	5.2	Visual
103	Nails or staples loose or missing		0.40%	5.2	Visual
104	Strapping missing or improperly placed		0.40%	5.2	Visual
MINOR					
201	Box blank cleats not fully seated		0.65%	5.2	Visual
202	Closing loops improperly fastened		0.65%	5.2	Visual
203	Strapping loose		0.65%	5.2	Visual/Manual
204	Marking incorrect or illegible		0.65%	5.3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

PRECAUTION WARNING

This specification covers sampling and testing toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.3.1 Sampling. The following representative samples shall be obtained from each batch of solid and liquid components using the procedures given in ASTM E300:

a. Solid Component - approximately 16.3 Kg (36 lbs) (one container).

b. Liquid Component - approximately 29 Kg (64 lbs) (two containers.

Sampling shall be in accordance with MIL-STD-1235, CSP-1 Plan, using a sample frequency code letter mutually agreed upon by the Contractor and the Contracting Officer. The samples shall be subjected to the tests listed in 4.4.3.1. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected. All batches produced between the time that the batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in para. 4.5. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch, the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.3.2	Demolition Kit XM268, blasting agent				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liquid component				
	Composition		100%	3.2	4.5.1.1
	pH		100%	3.3	4.5.1.2
	Powder component				
	Composition		6.5*	3.2	4.5.2.1
	Total volatiles		100%	3.4	4.5.2.2
	Mixed blasting agent				
	Density		100%	3.5	4.5.3.1
	Viscosity		100%	3.6	4.5.3.2
	Functioning		6.5*	3.7	4.5.3.3
	Detonation velocity		6.5*	3.8	4.5.3.3
	Workmanship		100%	3.10	4.5.4

NOTES: Samples selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, AQL 6.5 using as sample frequency code letter mutually agreed upon by the contractor and the contracting officer.

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4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes. (See 6.3)

4.5 Test methods and procedures (see 6.4). The tests in the subparagraphs of 4.5 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks. Also, unless otherwise specified herein, all chemicals and reagents shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Liquid component

4.5.1.1 Composition

4.5.1.1.1 Sodium perchlorate. Accurately weigh approximately 1.00 g of the liquid component in a tared 25 ml beaker. Transfer, quantitatively, to a one liter volumetric flask and dilute to the mark with distilled water. Take a one ml aliquot and transfer to a 250 ml volumetric flask and dilute to the mark with distilled water. Determine the sodium content by atomic absorption spectroscopy. Calculate the sodium perchlorate content as follows:

$$\text{Percent sodium perchlorate} = \frac{133.1A}{W}$$

Where: A = the ppm sodium determined by atomic absorbtion.
W = the weight in grams.

4.5.1.1.2 Ethylene glycol. Prepare calibration standards for the determination of ethylene glycol by accurately weighing 0.4500g, 0.5000 g and 0.5500 g of ethylene glycol into separate 5 ml volumetric flasks. Add 250 microliters of GC grade isopropyl alcohol to each volumetric flask with an appropriate syringe and again accurately weigh each flask. Fill each flask to the mark with distilled water and shake vigorously to thoroughly mix the solutions.

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Using the standard solutions determine the appropriate factor for the concentration of ethylene glycol vs the area under its gas chromatographic peak using the following instrument conditions (suggested):

Injection - 3 microliters
Column - 6 ft 1/8 inch stainless steel packed with 80/100 mesh Porapak Q
Carrier gas - Helium
Flow rate - 22cc per minute
Temperatures
Injection port - 235°C
Detector (Thermal Conductivity) - 230°C
Column (Programed) - 180° - 250°C at 10°C per minute with 8 minute hold at 250°C
Detector current - 200 milliamps

(The gas chromatograph should be equipped with an injection port liner in order to prevent the accumulation of potassium perchlorate crystals which will damage the column. The injection port liner should be removed and cleaned out after every tenth determination).

Calculate the response factor for each of the three ethylene glycol standard solutions each individually prepared standard sample using the following formula:

$$F = \frac{A \times B}{C \times D}$$

F = The ethylene glycol response factor
A = Ethylene glycol area count
B = Micrograms of isopropyl alcohol in standard solution
C = Micrograms of ethylene glycol in standard solution
D = Isopropyl alcohol area count

To obtain the ethylene glycol response factor take the average of the values obtained for the standard samples containing 0.4500g, 0.5000 g and 0.5500 g ethylene glycol.

To determine the ethylene glycol content of the liquid component place approximately 5 g in a tared 5 ml volumetric flask and weigh accurately. Add 250 microliters of GC grade isopropyl alcohol to the volumetric flask with a syringe and again accurately weigh. Dilute the mixture in the volumetric flask with distilled water to the 5.00cc mark. Shake the volumetric flask vigorously to thoroughly

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mix the solution. Inject 3 microliters into the gas chromatograph using the same conditions as were used for the calibration standard samples.

Determine the ethylene glycol content of the liquid component using the following formula:

$$A = \frac{B \times C \times 100}{D \times F \times W}$$

Where: A = Percent ethylene glycol in the liquid component
B = Area count for ethylene glycol peak in sample
C = Weight of isopropyl alcohol added to the liquid component sample
D = Area count for isopropyl alcohol peak in sample
F = Average response factor calculated for the three calibration standard samples
W = The weight of liquid component sample

4.5.1.1.3 Water. The percent water will be determined using the following formula:

$$A = 100 - (B + C)$$

Where: A = Percent water
B = Percent sodium perchlorate as determined in 4.5.1.1.1
C = Percent ethylene glycol as determined in 4.5.1.1.2

4.5.1.2 pH. The pH of the liquid component shall be measured using an electronic pH meter in the manner recommended by the instrument manufacturer.

4.5.2 Powder component

4.5.2.1 Composition

4.5.2.1.1 Aluminum powder coated with isostearic acid. Transfer an accurate weight of appproximately 1.00 g of the sample to a 30 cc tared medium porosity fritted glass crucible with stirring rod. Place crucible on an extraction filtrator and extract with 6 cc of diethyl ether. Repeat for a total of four extractions. Dry at 65°C. Cool and weigh. Calculate the loss in weight as isostearic acid using the following formula:

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$$A = \frac{100\ (B - C - WT)}{W - WT}$$

Where: A = Percent isostearic acid
B = Weight of sample and crucible before extraction, g
C = Weight of residue and crucible after extraction, g
W = Weight of sample (g)
T = Total volatiles content of the sample expressed as a decimal (see 4.5.2.2)

Extract the sample from the isostearic acid determination with 4 cc of warm water. Stir vigorously for 5-10 seconds and apply the vacuum while continuing to stir vigorously. Repeat the extraction without discontinuing the vacuum. Change the receiver, after the water extraction and wash the sample thoroughly 3 or 4 times with vigorous stirring with 95% ethyl alcohol (NOTE: This procedure is found to be best to prevent clogging of the crucible. In the event clogging occurs, fill the crucible with 95% ethyl alcohol, apply vacuum and stir vigorously). Dry and weigh. Reserve the water extract for determination of potassium acid phthalate.

Calculate the percent aluminum using the following formula:

$$D = \frac{100\ (E - F)}{W - TW}$$

Where: D = The percent aluminum
E = The weight of crucible and residue after water extraction, g
F = The weight of the crucible, g
W = The sample weight
T = The total volatiles content of the sample expressed as a decimal. See 4.5.2.2)

Calculate the aluminum coated with isostearic acid as follows:

$$G = A + D$$

Where: G = The percent aluminum and isostearic acid
A = The percent isostearic acid
D = The percent aluminum

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4.5.2.1.2 Guar gum. Prepare the following solutions for use in the analysis of guar gum content in the powder component: 4% resorcinol - 1.0 g resorcinol in 24 ml distilled water (prepare fresh daily), guar gum - accurately weigh 0.100 g of guar gum and disolve in distilled water using a mechanical stirer, transfer to a 100 ml volumetric flash and dilute to the mark with distilled water, potassium acid phthalate - dissolve 0.150 g of potassium acid phthalate in distilled water in a 100 ml volumetric flash and dilute to the mark with distilled water, isostearic acid - place 0.070 g of isostearic acid in a 100 ml volumetric flash and dilute to the mark with ethyl ether and a 20% sulfuric acid solution.

Determine the guar gum content as follows: Weigh accurately approximately 0.150 g of sample and quantitatively transfer to a 100 ml beaker. Add 10 ml of 20% sulfuric acid and heat gently until the aluminum powder disappears. Remove from heat, cool and dilute to 30-35 mls with distilled water. (If salt precipitates, heat slightly to redissolve). Filter thru a Whatman No. 40 (9 cm) filter into a 100 ml volumetric flask. Wash several times with 5 ml portions of distilled water. Dilute to the mark and mix thoroughly. Add to a 25 ml Erlenmeyer flask, in the order prescribed (using pipettes) 1 ml of the sample, then 1 ml of 4% resorcinol and 6 ml of concentrated sulfuric acid. Swirl the flask to mix thoroughly, then place in the ice bath until the flask temperature is reduced to room temperature or below. Remove from bath and allow to stand 15-20 minutes for color to develop. Read the absorbance of the sample against a dilute solution of sulfuric acid (2:6) using 1 cm cells at 416 millimicrons with a Beckman Model DU spectrophotometer or equal.

Determine the factor for the spectrophotometric analysis as follows: Prepare a 0.01 mg standard of guar gum as follows: Add approximately 1 ml of the isostearic solution to a 100 ml beaker. Remove the ether by a stream of dry air. Weigh approximately 0.146 g of the aluminum powder and transfer to the beaker. Add approximately 1 ml of the potassium acid phthalate solution and then accurately transfer 1 ml (1.0 mg) of the guar gum solution to the beaker. Add 10 ml of 20% sulfuric acid and proceed as with the sample.

Prepare and process in a like manner, at least two additional standards, one containing 2.0 ml (2.0 mg) of the gum, the second 3.0 ml (3.0 mg).

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Prepare a blank sample containing aluminum, isostearic acid and potassium acid phthalate in the same proportions as were used in the preparation of standards. Process this blank sample in the same manner as the sample and standards. The absorbance obtained for the blank sample shall be subtracted from the absorbances obtained for the sample and standards. Calculate the factor (F) for use in determining guar gum content as follows:

$$F = \frac{C}{A}$$

Where: C = Concentration of guar gum contained in the standard
and A = The absorbance obtained for the standard sample less the absorbance obtained for the blank.

Average the F values obtained for all of the standard determinations and use this value to determine the guar gum content of the powder component as follows:

$$G = \frac{F \times A \times D}{W}$$

Where: G = The guar gum content in percent
F = The factor obtained for the guar gum standards
A = The absorbance obtained for the sample less the absorbance obtained for the blank
D = The dilution factor for the sample (D = 100 for sample diluted in the 100 ml volumetric flask)
and W = The sample weight in milligrams.

4.5.2.1.3 Potassium acid phthalate. Transfer the water extract reserved from the aluminum determination quantitatively, to a liter volumetric flask. Dilute to the mark. Take a 25 ml aliquot and transfer to a 100 ml volumetric flask. Dilute to the mark and determine the potassium content by atomic absorption spectroscopy. Calculate the percent potassium acid phthalate by:

$$\frac{\text{ppm potassium} \times 2.09}{\text{wt}} = \% \text{ potassium acid phthalate}$$

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4.5.2.2 Total volatiles. The total volatiles of the powder component shall be determined by placing an accurately weighed 25 g portion of the powder in an aluminum weighing dish, which is provided with a cover lid, such that the depth of the powder does not exceed 0.4 inch. The aluminum dish and its cover lid shall have been previously dried and tared. With the cover lid removed place the aluminum dish and contents in an oven maintained at 70 ± 2°C for 4 hours. Remove the dish and its contents and allow to cool to room temperature in a desiccator. Replace the cover lid and weigh the aluminum powder and container. Calculate the total volatiles of the powder component using the following equation:

$$V = \frac{(A-B)\ 100}{W}$$

Where: A = Weight of dish cover and contents prior to drying.
B = Weight of dish cover and contents after drying.
W = Weight of the sample prior to drying.

4.5.3 Mixed blasting agent

4.5.3.1 Density. The density of the mixed blasting agent shall be determined using method T510.3, MIL-STD-286. The blasting agent shall be mixed for 15 minutes prior to testing.

4.5.3.2 Viscosity. Condition liquid and powder components for a minimum of 24 hours at temperature of 25°C. Add slowly 72 grams powder component to 128 grams liquid component while stirring vigorously by hand. Continue stirring the mixture intermittently for 15 minutes. There should be no evidence of lumps which would indicate improperly dispersed gum. The viscosity is determined immediately by using a Brookfield Synchro-Lectric Viscometer, Model HAT, running a 50 RPM with No. 4 spindle.

4.5.3.3 Functioning test and detonation velocity. Prepare 36.3 Kg (80 lbs) of blasting agent by thoroughly mixing the liquid and powder components in the proportions given in 3.2. Determine its density using the method given in 4.5.3.1. If the density value of the blasting agent meets the requirement given in 3.5, transfer the sample to two 4 inch by 24 inch long cardboard tubes which have been sealed at the bottom and fitted with probes for determining the detonation velocity. Each probe shall consist of

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two copper wires with plastic insulation. The wires shall be doubled back so that the bare portion of the wire is outside the cardboard tube. One pair of probes shall be inserted through small holes in the cardboard tube at a height of 14 inches above the bottom. The second pair at a height four inches above the bottom of the tube so that the pairs of probes are ten inches apart. The probes shall be connected to an interval timer with a microsecond read-out so that breaking the first pair of probes starts the timer and breaking the second pair stops it. Allow the samples to equilibrate for four hours at -30 $\pm$ 2°C. Place one of the cardboard containers on a 6 inch square by 1/2 inch thick mild steel (SAE 1010 to 1030) witness plate. Observing all appropriate safety procedures place a 1/2 pound TNT, or TNT base composition, booster charge at the open end of the cardboard tube partially submerged in the blasting agent sample. Initiate the booster charge with primer cord and/or a blasting cap. (The use and initiation of booster charges (demolition blocks) and initiating components (blasting caps, detonating cord, etc.) shall be in accordance with FM5-25 paragraph 2.1). The blasting agent shall be considered acceptable if there is no undetonated portion of either cardboard tube remaining and the witness plate is fractured.

Determine the detonation velocity using the following formula:

$$D = \frac{254}{T}$$

Where: D = The detonation velocity in millimeters/microseconds
T = The time from the interval timer in microseconds

Repeat the test with two additional samples allowing them to first equilibrate for four hours at 52 $\pm$ 2°C.

4.5.3.4 Bullet impact sensitivity. Twenty non-confining cylindrical containers approximately 4 inches in diameter and 6 inches in height shall be filled with freshly prepared blasting agent which has been mixed to meet the minimum density requirement given in 3.5. The sample shall be equilibrated for one hour at 25 $\pm$ 2°C. Observing all appropriate safety precautions place one of the containers 30.48 meters (100 ft) from a 30 caliber

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rifle and fire a 30 caliber projectile perpendicularly into the container. The muzzle velocity of the projectile shall be at least 853.4 meters (2800 ft) per second. The procedure shall be repeated with the remaining samples.

4.5.4 Workmanship. Each batch of the blasting agent shall be observed during processing and testing for evidence of foreign material or contamination.

4.5.5 Leakage of sealed container. The container filled with liquid or solid blasting agent component shall be placed in a compression chamber having prior approval of the Contracting Officer. The compression chamber shall be fitted with a pressure gauge having such range and sensitivity that a change of 1/8 pound per square inch (psi) can be easily read. The chamber shall be pressurized to 3 to 5 psi with air within two seconds and the air source turned off. The filled container shall be considered acceptable if the pressure does not change within 8 sec (min).

5. PACKAGING

5.1 Packaging - Level A

5.1.1 Liquid and solid blasting agent components. Liquid and solid blasting agent components shall be packaged in containers meeting the requirements of MIL-C-63407 and in accordance with drawing 11828917. Sealed containers shall be leak tested as described in 4.5.5.

5.1.2 Charge bags. Charge bags for blasting agent shall be manufactured in accordance with dwg 11828919. Packaging of the charge bags shall be in accordance with dwg 11828915.

5.1.3 Bag ties. Bag ties for the blasting shall be manufactured in accordance with dwg 9329739. Packaging of the bag ties shall be in accordance with dwg 11828915.

5.1.4 Goggles. Goggles manufactured in accordance with dwg 9328450 shall be packaged in accordance with dwg 11828915.

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5.1.5 Rubber gloves. Rubber gloves manufactured in accordance with dwg 9328450 shall be packaged in accordance with dwg 11828915.

5.1.6 Technical manual. Technical manual, Draft Equipment Publication for Demolition Kit, Blasting: XM268 (Blasting Agent), DEP 9-1375-529-14 (DRSAR-MAD-I), shall be packaged in accordance with dwg 11828915.

5.2 Packing Level A. The Demolition Kit shall be packed in accordance with dwg 11828915.

5.2.1 Packing Level B. Empty plastic containers, caps, seals, and Blasting Agent bags, ties, goggles and rubber gloves shall be packed for separate shipment in accordance with standard MIL-STD-1169.

5.3 Marking

5.3.1 Container, assembly. Container assemblies for the liquid and solid components shall be marked in accordance with dwg 11828917.

5.3.2 Blasting agent bags. The barrier bag containing the bags for blasting agent and ties shall be marked in accordance with dwg 11828915. The ink used shall be a permanent black ink that is resistant to smearing.

5.3.3 Goggles. The barrier bag for the goggles shall be marked in accordance with dwg 11828915.

5.3.4 Rubber gloves. The barrier bags containing the rubber gloves shall be marked in accordance with dwg 11828915.

5.3.5 Technical manual. The barrier bag containing the technical manual shall be marked in accordance with dwg 11828915.

6. NOTES

6.1 Intended use. The blasting agent covered by this specification is intended to be used for demolition and to create anti armor obstacles.

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6.2 Ordering data. See MIL-A-48078 (PA).

6.3 Submission of inspection equipment designs for approval See MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAR, Dover, New Jersey 07801.

6.4 Equivalent methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

Custodian:
ARMY-AR

Preparing Activity:
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